NAMANGA SECONDARY SCHOOL

FORM FOUR

INFORMATION AND COMPUTER STUDIES

DATABASE PRACTICAL EXAMINATION

School-Based Assessment / Mock Practical

TIME: 3 HOURS TOTAL: 100 MARKS

CANDIDATE DETAILS	
Candidate Name	_____
Candidate Number	_____
Class / Stream	_____
Date	_____

INSTRUCTIONS TO CANDIDATES

1. Use Microsoft Access to do all tasks.
2. Create the database and save it as NAMANGA_STUDENT_DATABASE.
3. Create the three tables: STUDENT, CLASS and ACTIVITY.
4. Enter only the records given in this paper.
5. Create only the relationship stated in Question C.
6. Use Design View, Query Design, Lookup Wizard, forms and reports where required.
7. Save every object with the exact name given in the question.
8. Check your work before submitting the database.
9. Answer all questions.

CASE STUDY

Namanga Secondary School wants a small database to keep student, class and school activity information. Create the database and use it to find useful information through queries, forms and reports.

IMPORTANT

The three tables are independent except for the CLASS–STUDENT relationship required in Question C. Do not create any relationship involving ACTIVITY.

QUESTION A — DATABASE CREATION

(a). Create a new database and save it as NAMANGA_STUDENT_DATABASE. (2 marks)

(b). Create the following three tables: STUDENT, CLASS and ACTIVITY. (3 marks)

QUESTION B — TABLE DESIGN AND DATA ENTRY

Create the tables in Design View. Choose the data types shown below, set the primary key for each table, and enter the records.

Table 1: STUDENT

StudentID	FirstName	LastName	Gender	DOB	ClassCode	Phone	Mark
ST001	Amani	Mollel	Male	12/03/2010	F2A	0712345678	78
ST002	Neema	Joseph	Female	27/07/2011	F2A	0756789012	84
ST003	Musa	Kweka	Male	19/01/2009	F3A	0765432109	67
ST004	Halima	Said	Female	05/11/2010	F2B	0687654321	72
ST005	Baraka	Mushi	Male	22/09/2008	F3A	0711223344	91
ST006	Maria	John	Female	14/02/2012	F1A	0744556677	58
ST007	Kelvin	Mrema	Male	30/06/2011	F2B	0788990011	63
ST008	Rehema	Nnko	Female	17/08/2009	F3B	0622334455	88
ST009	Omari	Juma	Male	02/05/2010	F2A	0699887766	49
ST010	Asha	Mwakalinga	Female	25/12/2008	F3B	0714556677	76

Primary key: StudentID

Table 2: CLASS

ClassCode	ClassName	Stream	RoomNo	ClassTeacher
F1A	Form I	A	R01	Mr Kato
F2A	Form II	A	R02	Ms Neema
F2B	Form II	B	R03	Mr Baraka
F3A	Form III	A	R04	Ms Asha
F3B	Form III	B	R05	Mr Musa

Primary key: ClassCode

Table 3: ACTIVITY

ActivityID	ActivityName	Category	ActivityDate	Participants	Fee
AC01	Cyber Safety Talk	ICT	20/01/2026	80	0
AC02	Football Match	Sports	24/01/2026	96	500
AC03	Database Club	Academic	28/01/2026	24	0
AC04	Science Fair	Academic	13/02/2026	55	1000
AC05	Girls in ICT Day	ICT	28/02/2026	45	0
AC06	Netball Match	Sports	07/03/2026	60	300
AC07	ICT Workshop	ICT	21/03/2026	38	2000
AC08	Programming Challenge	ICT	28/03/2026	27	500

Primary key: ActivityID

Total for Question B: 20 marks

QUESTION C — RELATIONSHIP

(a). Create a relationship between CLASS.ClassCode and STUDENT.ClassCode. (3 marks)

(b). Make ClassCode in CLASS the primary key and ClassCode in STUDENT the foreign key. (2 marks)

(c). Select Enforce Referential Integrity. (2 marks)

(d). Do not create a relationship for ACTIVITY. (1 marks)

The relationship should be one-to-many (1:M): one class can have many students.

QUESTION D — VALIDATION AND LOOKUP

(a). Use Lookup Wizard for STUDENT.Gender. The values must be Male and Female. Do not allow other values. (2 marks)

(b). Use Lookup Wizard for STUDENT.ClassCode. Get the values from the CLASS table. (2 marks)

(c). Set the validation rule for Mark to Between 0 And 100. (2 marks)

(d). Set Phone as Short Text and allow only 10 digits starting with 0. (1 marks)

(e). Set the validation rule for Participants to >=1. (1 marks)

(f). Set the validation rule for Fee to >=0. (1 marks)

(g). Set the validation rule for ActivityDate to dates in the year 2026 only. (1 marks)

Use the exact object names and rules requested. Check each rule by entering a wrong value and seeing whether Access rejects it.

QUESTION E — QUERIES

Create the queries using Query Design. Save each query with the exact name given.

No.	Task	Marks
1	Show StudentID, FirstName, LastName and Mark for all students. Save as Q_AllStudents.	1
2	Show students whose FirstName starts with M. Use Like "M*". Save as Q_NameStartsM.	1
3	Show students whose LastName contains ar. Use LIKE. Save as Q_LastNameAR.	1
4	Show students whose ClassCode is F2A, F2B or F3A. Use IN. Save as Q_ClassIN.	1
5	Show students with Mark >=75. Save as Q_Mark75Plus.	1
6	Show students with Mark Between 60 And 80. Save as Q_Mark60_80.	1
7	Show female students with Mark >=70. Use AND. Save as Q_Female70Plus.	2
8	Show students who are Male OR have Mark >=90. Save as Q_MaleOR90.	2
9	Show students born between 01/01/2009 and 31/12/2010. Save as Q_DOB_2009_2010.	2
10	Show students born before 01/01/2010 AND Mark >=70. Save as Q_Older70Plus.	2
11	Create an Age field and show students aged 16 or more. Save as Q_Age16Plus.	3

QUESTION E — QUERIES (CONTINUED)

No.	Task	Marks
12	Use STUDENT and CLASS to show FirstName, LastName, ClassName, Stream and Mark for Form III students. Save as Q_FormIII.	2
13	Create PerformanceBand: 80+ = Excellent, 60–79 = Good, below 60 = Needs Improvement. Save as Q_Performance.	3
14	Create a parameter query that asks the user to enter the minimum Mark. Save as Q_MinimumMark.	2
15	Create a crosstab showing the number of students by ClassName and Gender. Save as Q_ClassGender.	2
16	Show activities with Participants >=50 AND Fee <=1000. Save as Q_LargeLowCost.	1
17	Show activities from 01/02/2026 to 31/03/2026 where Category is ICT OR Academic. Use AND and OR. Save as Q_ICT_Academic.	2
18	Show activities whose name starts with C OR ends with Day. Save as Q_ActivityPattern.	1
19	Create TotalCollection = Participants × Fee. Show records where TotalCollection is >50000. Save as Q_TotalCollection.	2
20	Show activities with Participants Between 30 And 80 OR Fee = 0. Save as Q_ParticipantsORFree.	1

Total for Question E: 30 marks

QUESTION F — SORTING AND FILTERING

(a). Sort students by Mark from highest to lowest and show the top five. Save as Q_TopFive. (2 marks)

(b). Show only students in F2A and F2B. Save as Q_FormII. (1 marks)

(c). Sort ACTIVITY by ActivityDate from earliest to latest. Save as Q_ActivityDate. (1 marks)

(d). Sort ACTIVITY by Participants from highest to lowest. Save as Q_MostParticipants. (1 marks)

QUESTION G — FORMS

(a). Create FORM_STUDENT from STUDENT. Use a Columnar layout. (2 marks)

(b). Create FORM_CLASS from CLASS. Use a Tabular layout. (2 marks)

(c). Create FORM_ACTIVITY from ACTIVITY. Use a Justified layout. (2 marks)

QUESTION H — REPORTS

(a). Create REPORT_STUDENT_CLASS using Q_FormIII. Group by ClassName and sort Mark from highest to lowest. (2 marks)

(b). Create REPORT_ACTIVITY_CATEGORY using ACTIVITY. Group by Category and show ActivityName, Participants and Fee. (2 marks)

(c). Create REPORT_PERFORMANCE using Q_Performance. Show student name, Mark and PerformanceBand. Add a title and page number. (2 marks)

QUESTION I — TESTING

Test the validation rules using the values below. Each test must give the stated result.

No.	Test value	Expected result
1	Mark = 105	Reject
2	Participants = 0	Reject
3	Fee = -500	Reject
4	Gender = Other	Reject
5	Phone = 07123456	Reject
6	ClassCode = F9A	Reject because of the relationship

After testing, return the database to valid data before saving it.

QUESTION J — DATABASE CHECK

(a). Check that the three tables, one relationship, all queries, three forms and three reports are saved. (1 marks)

(b). Make sure all object names are correct. (1 marks)

QUESTION K — FINAL SUBMISSION

(a). Preview the three reports and make sure no information is cut off. (1 marks)

(b). Save and submit the final database as instructed by the teacher. (1 marks)

END OF EXAMINATION

MARK SUMMARY — FOR EXAMINER USE

Section	Description	Marks
A	Database creation	5
B	Table design and data entry	20
C	Relationship	8
D	Validation and Lookup	10
E	Queries	30
F	Sorting and filtering	5
G	Forms	6
H	Reports	6
I	Testing	6
J	Database check	2
K	Final submission	2
TOTAL		100

Candidate's Total: _____ / 100

Examiner's Name: _____ Signature: _____

Date: _____